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Software Requirements Specification

E-Poch Medical System

Clinic & Dispensary Management Platform — Software Requirements Specification

Modules: Authentication • Patient • Family • Appointment • Consultation • Prescription • Pharmacy • Inventory • Supplier
• Billing • Reports • Dashboard • Settings

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1.0	Original	Clinic IT Committee	Initial brief requirements outline
2.0	29 July 2026	Business Analysis Team	Expanded to full SRS: use cases, diagrams, data dictionary, API & security specification
3.0	30 July 2026	Business Analysis Team	Rebranded as E-Poch Medical System; added Section 10 – Technology Stack & Architecture
4.0	30 July 2026	Business Analysis Team	Revised Section 10: frontend changed to React (SPA), backend changed to Node.js/Express, replacing the earlier unified NestJS approach

Approval

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1. Executive Summary

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for E-Poch Medical System, a web-based Clinic & Dispensary Management System intended for a private doctor's practice that also operates an in-house pharmacy. The system will digitize the full patient journey — from registration and appointment scheduling through consultation, electronic prescribing, medicine dispensing, billing, and reporting — replacing manual registers and spreadsheet-based tracking.

The platform is designed around five core actors: Administrators, Receptionists, Doctors, Pharmacists, and (in a future phase) Laboratory staff. It enforces role-based access control, maintains an auditable trail of every clinical and financial transaction, and applies pharmacy-grade inventory controls such as First-Expiry-First-Out (FEFO) batch allocation, expiry alerting, and automatic stock deduction on dispense.

This document expands the original one-page requirements brief into a complete specification suitable for design, development, quality assurance, and stakeholder sign-off. It includes over 100 discrete functional requirements, detailed use cases, user stories, activity and sequence diagrams, an entity-relationship model, a full database dictionary, UI and API specifications, a dedicated technology stack & architecture specification, security requirements, and acceptance criteria.

2. Problem Statement

The clinic currently manages patient registration, consultation notes, prescriptions, pharmacy stock, and billing using a combination of paper registers, handwritten prescriptions, and disconnected spreadsheets. This manual approach creates the following operational problems:

- Duplicate and fragmented records — the same patient may be registered multiple times under slightly different name spellings, with no reliable way to detect duplicates via National Identity Card (NIC) number.
- No family-linked view — receptionists cannot quickly see which patients belong to the same household, complicating billing and history lookups for dependents.
- Slow, error-prone queueing — patient order at the doctor's room is tracked verbally or on paper, leading to disputes and lost time.
- Illegible or lost prescriptions — handwritten prescriptions are difficult for the pharmacist to read accurately and cannot be validated against real-time stock.
- Uncontrolled inventory — expired medicines are sometimes dispensed unintentionally; stock-outs are discovered only when a patient is already at the pharmacy counter.
- Manual, disconnected billing — consultation fees and medicine charges are calculated and recorded separately, increasing the risk of billing errors and revenue leakage.
- No analytics — the clinic has no consolidated view of patient volume, revenue trends, fast-moving medicines, or doctor productivity.
- Weak accountability — there is no audit trail showing who registered, edited, prescribed, dispensed, or billed a given transaction.

These problems reduce patient throughput, increase clinical risk (e.g., dispensing expired stock), and make it difficult for the clinic owner to make informed operational and purchasing decisions.

3. Project Objectives

#	Objective	Success Indicator
O1	Eliminate duplicate patient records	Zero duplicate NIC entries after go-live
O2	Reduce average patient waiting time	Queue-to-consultation time reduced by 30%
O3	Prevent dispensing of expired/insufficient	100% of dispense transactions pass automated

#	Objective	Success Indicator
	stock	FEFO & stock validation
O4	Provide real-time, low-stock and expiry visibility	Automated alerts generated before stock-outs occur
O5	Unify consultation and pharmacy billing	Single consolidated invoice per visit
O6	Enable data-driven decisions	Dashboards and exportable reports available to Admin/Doctor roles
O7	Ensure accountability and traceability	Every create/update/delete action is captured in the audit log
O8	Protect patient data	RBAC, encrypted passwords and HTTPS enforced clinic-wide

4. Scope

4.1. In Scope

- Authentication and role-based access control for Admin, Receptionist, Doctor and Pharmacist roles.
- Patient registration with duplicate NIC validation and system-generated unique Patient ID.
- Family management linking multiple patients under a single family unit.
- Appointment scheduling and real-time queue management (Waiting → Called → Consulting → Completed).
- Doctor consultation workspace showing history, allergies, vitals and clinical notes.
- Electronic prescription creation with medicine lookup and live stock validation.
- Pharmacy queue and dispensing workflow (Pending → Preparing → Dispensed → Collected).
- Drug and inventory management including batch tracking, FEFO allocation, expiry alerts and low-stock notifications.
- Supplier and purchase order management.
- Billing, invoice generation and payment recording.
- Patient history timeline consolidating visits, prescriptions and invoices.
- Reports and analytics dashboards with PDF/Excel export.

4.2. Out of Scope (Phase 1)

- Laboratory test ordering and results management (planned for a future phase, see Section 21).
- Online/public self-service appointment booking portal.
- Native mobile applications (a responsive web UI is in scope; native iOS/Android apps are not).
- WhatsApp / SMS reminder integrations.
- AI-assisted clinical decision support.
- Multi-branch / multi-clinic consolidated operation.

4.3. Assumptions and Constraints

- The clinic has stable internet connectivity at the point of care sufficient to run a web-based application in the cloud deployment profile; the on-premise profile (Section 10.5) is available where connectivity is not guaranteed.
- Each staff member will be issued an individual login; shared logins are not permitted for audit reasons.

- NIC number is available for the vast majority of adult patients; a fallback identifier (guardian NIC + DOB) is used for minors.
- The clinic will supply barcode/QR-capable hardware if barcode-based medicine scanning is enabled.
- Regulatory recordkeeping requirements for prescriptions and controlled substances follow local health authority guidelines and are configurable per jurisdiction.
- Where more than one workstation writes to the database concurrently, the cloud/PostgreSQL deployment profile is assumed rather than the single-file SQLite profile (see Section 10.5).

5. Stakeholders

Stakeholder	Interest / Role	Key Concerns
Clinic Owner / Doctor (Sponsor)	Owns the clinic; primary decision-maker and often also the treating doctor	ROI, patient safety, regulatory compliance, revenue visibility
Receptionist	Front-desk operations: registration, appointments, queue, initial billing	Speed of data entry, ease of duplicate detection
Doctor(s)	Conduct consultations, record clinical notes, issue prescriptions	Fast access to patient history, minimal clicks during consultation
Pharmacist	Manages inventory and dispenses medicine against prescriptions	Accurate stock levels, expiry safety, clear dispensing queue
Administrator / IT Support	System configuration, user management, backups, reporting	System reliability, security, ease of administration
Patients / Families	Indirect beneficiaries of faster, safer, more accurate service	Privacy of medical data, shorter waiting times, billing accuracy
Supplier / Vendor	Supplies medicines against purchase orders	Clear, timely purchase orders and payment records
Future Laboratory Staff	Will process lab test orders and results in a future phase	Integration readiness for orders and results exchange
Regulatory / Health Authority	External party with compliance interest in prescriptions and controlled substances	Accurate recordkeeping, data retention, audit trail

6. User Roles & Permissions

Access is governed by Role-Based Access Control (RBAC). Each user is assigned exactly one primary role; the Administrator may grant supplementary permissions where justified. The permission matrix below summarizes access at the module level (C = Create, R = Read, U = Update, D = Delete, X = Execute/Action).

Module	Admin	Receptionist	Doctor	Pharmacist
User & Role Management	CRUD	—	—	—
Patient Registration	CRUD	CRUD	R	R
Family Management	CRUD	CRUD	R	R
Appointments & Queue	CRUD	CRUD	RU (own queue)	R
Consultation Records	R	R (limited)	CRUD (own)	R
Prescriptions	R	R	CRUD (own)	R, X (dispense)

Module	Admin	Receptionist	Doctor	Pharmacist
Pharmacy Queue / Dispensing	R	R	R	CRUD, X
Inventory & Batches	CRUD	R	R	CRUD
Suppliers & Purchase Orders	CRUD	—	—	CR, RU
Billing & Invoices	CRUD	CRU	R	R, CU (pharmacy items)
Reports & Dashboards	Full access	Operational reports	Clinical/own reports	Inventory reports
System Settings & Audit Log	CRUD	—	—	—

6.1. Role Descriptions

- Administrator — Full system access. Manages users, roles, master data (medicines, suppliers), system settings, backups and audit logs.
- Receptionist — Registers patients and families, schedules appointments, manages the front-desk queue, and initiates billing.
- Doctor — Conducts consultations, records vitals/history/allergies/notes, and issues electronic prescriptions. Can view historical records of their own patients.
- Pharmacist — Manages the pharmacy queue, dispenses medicine against valid prescriptions, maintains stock batches, and raises purchase orders.
- Laboratory Staff (future role) — Reserved role for the future laboratory integration; not active in Phase 1.

7. Functional Requirements

Functional requirements are grouped by module and numbered sequentially (FR-001 through FR-091). Each requirement is tagged with priority: M = Must-have, S = Should-have, C = Could-have, following the MoSCoW prioritization method.

7.1. Authentication & Role-Based Access (FR1)

ID	Requirement	Priority
FR-001	Users shall log in with a unique username/email and password over an encrypted (HTTPS) connection.	M
FR-002	The system shall enforce role-based access control restricting each screen and API endpoint per the permission matrix in Section 6.	M
FR-003	Passwords shall be stored using a salted one-way hash (e.g., bcrypt/argon2); plaintext passwords shall never be persisted or logged.	M
FR-004	The system shall lock an account for a configurable cool-down period after 5 consecutive failed login attempts.	M
FR-005	The system shall support a self-service 'forgot password' flow via a time-limited reset link.	S

ID	Requirement	Priority
FR-006	Administrators shall be able to create, deactivate, and reassign the role of any staff user.	M
FR-007	The system shall automatically log out an idle session after a configurable timeout (default 15 minutes).	S
FR-008	The system shall support optional two-factor authentication (2FA) for Administrator accounts.	C

7.2. Patient Registration (FR2)

ID	Requirement	Priority
FR-009	The system shall capture patient demographic details: full name, NIC/passport number, date of birth, gender, phone, address, and blood group.	M
FR-010	The system shall validate NIC uniqueness at the point of entry and block registration if a duplicate NIC is detected.	M
FR-011	The system shall auto-generate a unique, human-readable Patient ID (e.g., PT-000123) upon successful registration.	M
FR-012	The system shall support registration of minors using guardian NIC and date of birth as an alternate unique key.	M
FR-013	The system shall allow capturing known allergies and chronic conditions at registration time.	S
FR-014	The system shall support editing patient demographic data with a full change history retained.	M
FR-015	The system shall support patient photo capture/upload for positive identification.	C
FR-016	The system shall allow searching patients by NIC, name, phone number, or Patient ID with results returned in under 2 seconds.	M
FR-017	The system shall flag possible near-duplicate patients (fuzzy name + DOB match) for receptionist review even when NIC differs.	S

7.3. Family Management (FR3)

ID	Requirement	Priority
FR-018	The system shall allow grouping multiple patients under a single Family record.	M
FR-019	Each patient shall belong to exactly one family; each family may contain many patients (one-to-many).	M
FR-020	The system shall designate a Head of Family for billing and contact purposes.	M
FR-021	The system shall allow viewing all members and combined visit/billing history of a family from one screen.	S
FR-022	The system shall support moving a patient from one family to another with an audit record of the change.	S

ID	Requirement	Priority
FR-023	The system shall support merging two family records where a duplicate family entry is discovered.	C

7.4. Appointment Scheduling & Queue Management (FR4)

ID	Requirement	Priority
FR-024	The system shall allow receptionists to book, reschedule, and cancel appointments against a doctor's available slots.	M
FR-025	The system shall prevent double-booking of the same doctor for overlapping time slots.	M
FR-026	The system shall support walk-in patients being added directly to the day's queue without a pre-booked slot.	M
FR-027	The system shall maintain queue status per the state machine: Waiting → Called → Consulting → Completed.	M
FR-028	The system shall display a live queue board visible to reception and doctor screens, auto-refreshing without manual reload.	S
FR-029	The system shall allow a doctor to recall or skip a patient in the queue with reason capture.	S
FR-030	The system shall compute and display estimated waiting time based on average consultation duration.	C
FR-031	The system shall notify reception when a consultation is marked Completed so billing can proceed.	M

7.5. Doctor Consultation (FR5)

ID	Requirement	Priority
FR-032	The system shall present the doctor with the patient's consultation history, allergies and chronic conditions at the start of a visit.	M
FR-033	The system shall allow capturing vital signs (blood pressure, temperature, pulse, weight, height, BMI auto-calculated) per visit.	M
FR-034	The system shall allow free-text and structured clinical notes (diagnosis, complaint, examination findings) per visit.	M
FR-035	The system shall allow attaching diagnosis codes (e.g., ICD-10) to a consultation record.	S
FR-036	The system shall allow the doctor to record follow-up instructions and a recommended review date.	S
FR-037	The system shall prevent editing of a finalized consultation record except by the original doctor or an Administrator, with all edits logged.	M
FR-038	The system shall allow attaching scanned documents or images (e.g., prior reports) to a consultation.	C

7.6. Electronic Prescription (FR6)

ID	Requirement	Priority
FR-039	The system shall provide a searchable medicine lookup (by name, generic name, or category) while creating a prescription.	M
FR-040	The system shall validate, in real time, that sufficient unexpired stock exists for each prescribed item before allowing submission.	M
FR-041	The system shall capture dosage, frequency, duration, and administration route for each prescribed medicine.	M
FR-042	The system shall warn the doctor of a documented patient allergy that conflicts with a selected medicine.	S
FR-043	The system shall allow the doctor to mark a prescription item as a repeat/refill of a previous prescription.	S
FR-044	The system shall generate a printable/PDF prescription document bearing doctor name, registration number, and clinic letterhead.	M
FR-045	The system shall route a submitted prescription automatically to the Pharmacy Queue with status Pending.	M
FR-046	The system shall prevent submission of a prescription containing an expired or discontinued medicine.	M

7.7. Pharmacy Queue & Dispensing Workflow (FR7)

ID	Requirement	Priority
FR-047	The system shall display incoming prescriptions in a pharmacy queue ordered by submission time.	M
FR-048	The system shall track dispensing status per the state machine: Pending → Preparing → Dispensed → Collected.	M
FR-049	The system shall allow the pharmacist to substitute a prescribed medicine with an equivalent, subject to doctor/administrator-configured rules.	S
FR-050	The system shall block the Dispensed transition if any line item fails stock or expiry validation.	M
FR-051	The system shall record the dispensing pharmacist, timestamp, and batch number against every dispensed item.	M
FR-052	The system shall allow partial dispensing of a prescription with the remaining items held as pending.	S
FR-053	The system shall generate a dispensing label with medicine name, dosage instructions, and patient name.	S

7.8. Drug & Inventory Management (FR8)

ID	Requirement	Priority
FR-054	The system shall maintain a master medicine catalog including generic name, brand name, form, strength, and unit of measure.	M

ID	Requirement	Priority
FR-055	The system shall track stock at batch level, capturing batch number, manufacture date, expiry date, and quantity received.	M
FR-056	The system shall allocate stock for dispensing using First-Expiry-First-Out (FEFO) logic by default.	M
FR-057	The system shall generate an expiry alert for any batch nearing expiry within a configurable threshold (default 90 days).	M
FR-058	The system shall generate a low-stock notification when on-hand quantity for a medicine falls below its configured reorder level.	M
FR-059	The system shall automatically deduct stock from the allocated batch immediately upon successful dispensing.	M
FR-060	The system shall prevent manual stock deduction below zero and require a documented reason for manual stock adjustments.	M
FR-061	The system shall support stock-take / physical count reconciliation with variance reporting.	S
FR-062	The system shall support barcode/QR scanning for rapid batch identification during goods receipt and dispensing.	S
FR-063	The system shall maintain a full stock ledger (movement history) per batch for audit purposes.	M

7.9. Supplier & Purchase Order Management (FR9)

ID	Requirement	Priority
FR-064	The system shall maintain a supplier master with contact details, payment terms, and supplied medicine catalog.	M
FR-065	The system shall allow creating a purchase order (PO) against a supplier, listing medicines, quantities and expected cost.	M
FR-066	The system shall track PO status through a workflow: Draft → Submitted → Partially Received → Received → Closed.	M
FR-067	The system shall auto-suggest reorder quantities based on medicines flagged as low-stock.	S
FR-068	The system shall record Goods Received Notes (GRN) against a PO, creating new stock batches automatically.	M
FR-069	The system shall flag discrepancies between ordered and received quantities for administrator review.	S

7.10. Billing, Invoices & Payments (FR10)

ID	Requirement	Priority
FR-070	The system shall generate a single consolidated invoice per visit combining consultation fee and dispensed medicine charges.	M
FR-071	The system shall support cash, card, and mobile-payment methods, with the ability to record partial and split payments.	M

ID	Requirement	Priority
FR-072	The system shall apply configurable discount rules (e.g., senior citizen, staff, insurance) at the invoice line or total level.	S
FR-073	The system shall generate a printable/PDF receipt immediately upon payment confirmation.	M
FR-074	The system shall track outstanding balances and support recording of later/partial settlements.	S
FR-075	The system shall support invoice void/refund with mandatory reason capture and administrator approval.	M
FR-076	The system shall reconcile daily cash/card totals against recorded payments for end-of-day closing.	S

7.11. Patient History Timeline (FR11)

ID	Requirement	Priority
FR-077	The system shall present a chronological timeline per patient covering appointments, consultations, prescriptions and invoices.	M
FR-078	The system shall allow filtering the timeline by date range or record type.	S
FR-079	The system shall allow authorized users to export a patient's history as a PDF summary.	S
FR-080	The system shall visually flag recurring or chronic conditions across multiple visits.	C

7.12. Reports & Analytics Dashboards (FR12)

ID	Requirement	Priority
FR-081	The system shall provide an Administrator dashboard summarizing daily patient volume, revenue, and top-dispensed medicines.	M
FR-082	The system shall provide a Doctor dashboard summarizing personal consultation counts and follow-up due lists.	S
FR-083	The system shall provide a Pharmacist dashboard summarizing low-stock items, expiring batches, and today's dispensing volume.	M
FR-084	The system shall support exporting any report or table view to PDF and Excel formats.	M
FR-085	The system shall support date-range filtering and comparison (e.g., this month vs. last month) on all dashboards.	S
FR-086	The system shall provide a configurable report scheduler to email periodic reports to Administrators.	C

7.13. System Administration & Settings

ID	Requirement	Priority
FR-087	The system shall allow Administrators to configure clinic profile details (name,	M

ID	Requirement	Priority
	address, logo, registration number) used on printouts.	
FR-088	The system shall allow Administrators to configure reorder levels, expiry alert thresholds, and consultation fee schedules.	M
FR-089	The system shall provide a searchable audit log capturing user, action, entity, and timestamp for all create/update/delete operations.	M
FR-090	The system shall support scheduled daily automatic backups with restore verification.	M
FR-091	The system shall allow Administrators to manage master data lists (medicine categories, payment methods, discount types).	S

Total functional requirements specified: 91 (FR-001 to FR-091), covering all thirteen suggested modules.

8. Non-Functional Requirements

Category	ID	Requirement
Performance	NFR-01	Patient and medicine search results shall return within 2 seconds under normal load (up to 50 concurrent users).
	NFR-02	Dashboard widgets shall render within 3 seconds for a 12-month data range.
Usability	NFR-03	The UI shall be responsive, supporting desktop, tablet, and mobile-browser viewports.
	NFR-04	Core clinical workflows (consultation, prescription) shall be completable within 5 primary screens/steps.
Security	NFR-05	All traffic shall be served exclusively over HTTPS/TLS 1.2+.
	NFR-06	Passwords shall be encrypted at rest using a salted hash; sensitive PII shall be encrypted at rest.
	NFR-07	Role-based access control shall be enforced at both UI and API layers.
Auditability	NFR-08	All create, update, and delete actions on clinical or financial records shall be captured in an immutable audit log.
Reliability	NFR-09	The system shall target 99.5% uptime during clinic operating hours.
	NFR-10	A full daily backup shall be taken automatically and retained for a minimum of 90 days.
Interoperability	NFR-11	The system shall expose a documented REST API to support future laboratory and mobile integrations.
Portability	NFR-12	Reports and documents shall be exportable in PDF and Excel formats.
Scalability	NFR-13	The architecture shall scale horizontally to support future multi-branch deployment without redesign.
Hardware Support	NFR-14	The system shall support barcode and QR code scanning peripherals for medicine identification.
Maintainability	NFR-15	The codebase shall follow a modular architecture aligned to the 13

Category	ID	Requirement
		functional modules to ease independent updates.
Accessibility	NFR-16	Key workflows shall meet WCAG 2.1 AA color-contrast and keyboard-navigation guidelines.
Data Portability	NFR-17	The data layer shall be provider-agnostic at the ORM level, allowing the same schema to target an embedded local database or a managed cloud database without application-code changes.

9. Business Rules

ID	Rule
BR-01	One patient belongs to exactly one family; one family may contain many patients.
BR-02	Duplicate NIC numbers are not permitted across the patient database.
BR-03	Medicines cannot be dispensed if the allocated batch is expired or if available stock is insufficient for the prescribed quantity.
BR-04	Inventory quantities update automatically and immediately after a dispensing transaction is confirmed.
BR-05	Patient queue status must progress strictly through: Waiting → Called → Consulting → Completed (no skipping without an explicit override and reason).
BR-06	Pharmacy dispensing status must progress strictly through: Pending → Preparing → Dispensed → Collected.
BR-07	Stock allocation for dispensing follows First-Expiry-First-Out (FEFO) unless a documented clinical override is recorded.
BR-08	A consultation record, once finalized, may only be amended by the originating doctor or an Administrator, and every amendment is logged.
BR-09	An invoice may not be voided without Administrator approval and a documented reason.
BR-10	A user account must be assigned exactly one primary role at any given time.

10. Technology Stack & Architecture

This section defines the confirmed implementation stack for E-Poch Medical System and is new to this document. It translates the functional and non-functional requirements above into a concrete architecture, and calls out any additional components needed to deliver those requirements that were not implied by the original one-page brief.

10.1. Architectural Style

The system is built as a two-tier web application: a React single-page application (SPA) for the frontend, and a Node.js/Express REST + WebSocket API for the backend, communicating over HTTPS/JSON. The two are built, versioned, and deployed as separate artifacts — the React app compiles to static assets; the Express app runs as a Node.js process — which keeps the presentation layer and the business/data logic cleanly separated. The thirteen functional modules listed on the cover page (Authentication, Patient, Family, Appointment, Consultation, Prescription, Pharmacy, Inventory, Supplier, Billing, Reports, Dashboard, Settings) are reflected as a matching folder-per-module structure on both sides: page/feature folders in React, and router/controller/service/repository folders in Express, satisfying the modular-architecture requirement in NFR-15.

10.2. Core Stack

Layer	Technology	Role in this Project
Runtime	Node.js (LTS)	JavaScript runtime for the backend API server; the frontend build is compiled to static assets that run in the browser.
Frontend Framework	React (built with Vite)	Delivers the responsive single-page application described in Section 17. Component-based structure maps naturally onto the screen list in that section (Consultation Workspace, Pharmacy Queue, Dispensing Screen, etc.).
Frontend Routing & Data	React Router for navigation; TanStack Query (React Query) for server-state fetching, caching and refetch-on-focus	Keeps each screen's data (patient search, live queue, inventory alerts) in sync with the API without hand-rolled state management.
Frontend Styling	Tailwind CSS, with a small shared component library	Implements the consistent navigation, color-coding convention (green/amber/red) and responsive layout required by Section 17.1.
Backend Framework	Node.js + Express	A minimal, unopinionated REST API server. Because Express does not impose a module system the way NestJS does, the codebase is organized as one feature folder per functional module (auth, patients, families, appointments, consultations, prescriptions, pharmacy, inventory, suppliers, billing, reports, settings), each exposing its own router/controller/service/repository files — preserving the modular structure required by NFR-15 by convention rather than framework enforcement.
Real-Time Updates	Socket.IO (server) with the Socket.IO client in React	Pushes queue-status and pharmacy-queue changes to open screens without polling (FR-028, UC-02).
ORM / Data Access	Prisma	A single Prisma schema is shared by both database targets below; Prisma Migrate manages versioned schema migrations for each environment.
Local / On-Premise Database	SQLite (file-based), accessed through Prisma	For a single-workstation or single-clinic offline-capable deployment. Zero external database server to install or maintain — the entire clinic database is one file.
Cloud Database	PostgreSQL, accessed through Prisma	For the hosted/cloud deployment profile, and a prerequisite for the future multi-branch scaling goal in NFR-13. Supports the concurrent multi-role writes (reception, doctor, pharmacist working simultaneously) that SQLite handles

Layer	Technology	Role in this Project
		poorly.

10.3. Data Layer — Local vs. Cloud

Prisma is used as the single ORM for both database targets. The same Prisma schema (models, relations, and the tables in Section 16) is reused across environments; only the datasource provider and connection string change.

Environment	Database Engine	Managed Via
Local / On-Premise	SQLite (single embedded database file)	Prisma Client + Prisma Migrate
Cloud / Hosted	PostgreSQL (managed instance)	Prisma Client + Prisma Migrate

10.4. Supporting Libraries & Services

The original brief did not specify supporting infrastructure beyond the core stack. The following are the additional pieces needed to actually satisfy the functional and security requirements already listed in Sections 7, 8 and 18, and are recommended additions to the build:

Concern	Recommended Component	Satisfies
Authentication	Passport.js (passport-jwt strategy) + jsonwebtoken, bcrypt/argon2 for password hashing	Implements FR-001–FR-008 and SEC-01–SEC-03.
Authorization	Custom Express middleware reading the authenticated user's role and checking it against the Section 6 permission matrix	Enforces RBAC server-side on every route (SEC-04), independent of anything enforced in the React UI.
Input Validation	Zod schemas validated in an Express middleware layer	Server-side sanitization/validation of every request body and query string (SEC-10).
Scheduled Jobs	node-cron	Drives expiry alerts, low-stock checks, nightly backups, and report-scheduler emails (FR-057, FR-058, FR-090, FR-086).
Background/Queue Processing	BullMQ (Redis-backed) — cloud profile only	Offloads PDF/Excel generation and email sending so the request thread is never blocked.
PDF Generation	PDFKit or Puppeteer (headless Chromium)	Prescriptions, receipts, dispensing labels, and patient-history exports (FR-044, FR-073, FR-079).
Excel Export	ExcelJS	Report and dashboard exports (FR-084, NFR-12).
File Uploads	Multer	Patient photos and scanned consultation attachments (FR-015, FR-038).
Barcode / QR	Server: bwip-js or JsBarcode for	Implements FR-062 and NFR-14.

Concern	Recommended Component	Satisfies
	label generation. Browser (React component): a camera-based scanning library (e.g., html5-qrcode) for goods-receipt and dispensing scans.	
Email Delivery	Nodemailer (SMTP) or a transactional email API	Password-reset links and scheduled report emails (FR-005, FR-086).
Caching / Sessions (cloud profile)	Redis	Session/token blocklist storage for horizontally-scaled instances, plus the BullMQ backend above.
API Documentation	swagger-jsdoc + swagger-ui-express	Auto-generates and serves the OpenAPI spec for the endpoints in Section 18, supporting the future lab/mobile integrations in NFR-11.
Structured Logging	Pino (via pino-http middleware)	Correlates request logs with audit-log entries for troubleshooting and SEC-06 traceability.
Configuration Management	dotenv + a small Zod-validated config module, per-environment .env files	Switches the Prisma datasource (SQLite vs. PostgreSQL) and other environment-specific values without code changes.
Security Headers	Helmet middleware	Sets CSP, HSTS and related headers (SEC-13).
Rate Limiting	express-rate-limit	Protects login and public endpoints from brute-force/DoS abuse (SEC-08, SEC-14).
Testing	Jest + Supertest (backend API tests); React Testing Library + Vitest (frontend component tests)	Covers business rules such as FEFO allocation, NIC de-duplication, and RBAC middleware, plus critical UI flows.
Containerization	Docker + Docker Compose	Packages the Express API, the built React static assets, and (in the cloud profile) PostgreSQL and Redis, as reproducible services.
Reverse Proxy / TLS Termination	Nginx (or a managed load balancer in the cloud profile)	Serves the built React assets, proxies /api requests to Express, terminates HTTPS/TLS 1.2+, and redirects HTTP → HTTPS (SEC-01, NFR-05).
CI/CD	GitHub Actions (build → test → lint → deploy)	Builds the React bundle, runs the Express test suite, automates the Prisma migration step, and builds the Docker images on every release.
Backup Automation	Scheduled job copying/encrypting the SQLite file, or pg_dump for	Implements FR-090, NFR-10 and SEC-09.

Concern	Recommended Component	Satisfies
	PostgreSQL, to encrypted off-instance storage	

10.5. Deployment Topologies

Two deployment profiles are supported by the same codebase and Prisma schema:

	Profile A — On-Premise (SQLite)	Profile B — Cloud (PostgreSQL)
Target environment	Single workstation or single-LAN clinic with limited/unstable internet	Hosted cloud environment; also the required base for future multi-branch operation
Database	SQLite file via Prisma (embedded, no separate DB server)	PostgreSQL (managed instance) via Prisma
App process	One Express process serves the API; the React build is served as static files from the same process or a local Nginx	Containerized Express API instance(s) behind Nginx/load balancer, horizontally scalable per NFR-13; React build served as static assets via Nginx or a CDN
Background jobs / cache	In-process node-cron only (no Redis required)	node-cron + BullMQ + Redis for queued jobs and caching
Backups	Scheduled encrypted copy of the SQLite file to external/removable storage	Scheduled encrypted pg_dump to off-site/object storage, retained ≥ 90 days
Best suited to	Pilot rollout / a single clinic branch validating the system before wider rollout	Production rollout, multiple concurrent staff, and any future multi-branch phase

10.6. Implementation Notes & Risks

- **Concurrency caveat:** SQLite serializes writes at the file level. It is adequate for the on-premise, single-workstation profile described above, but if reception, the doctor, and the pharmacist will be writing to the database concurrently from separate machines, enable Write-Ahead Logging (WAL) mode at minimum, and treat the PostgreSQL/cloud profile as the recommended default for any clinic with more than one simultaneous user station.
- **Schema portability:** Prisma's schema is written once and targets either provider through the DATABASE_URL/datasource setting; a small number of Prisma field types (e.g., native enums, certain JSON operations) behave differently between SQLite and PostgreSQL and should be validated in both environments during QA, not assumed identical.
- Because Express itself does not enforce module boundaries the way NestJS does, the module boundaries described throughout Section 7 (Authentication, Patient, Family, Appointment, Consultation, Prescription, Pharmacy, Inventory, Supplier, Billing, Reports, Dashboard, Settings) must be maintained by team convention — one feature folder per module, each with its own router/controller/service/repository — to keep the "modular architecture" required by NFR-15.
- The React frontend and Express backend are separate deployable artifacts (a static asset bundle and an API server, respectively), unlike a unified server-rendered app; CORS configuration and a shared API contract (kept in sync via the OpenAPI spec in Section 18) become part of the build and release process.

11. Complete Use Cases

This section documents the primary use cases of the system in full IEEE-style format: actors, preconditions, main success scenario, alternate/exception flows, and postconditions.

UC-01: Register a New Patient

Actors	Receptionist
Preconditions	Receptionist is logged in with valid session.
Trigger	A new (previously unregistered) patient arrives at the clinic.

Main Success Scenario

1. Receptionist selects 'New Patient Registration'.
2. Receptionist enters NIC number; system checks for duplicates.
3. System confirms NIC is unique and enables the remaining fields.
4. Receptionist enters name, date of birth, gender, phone, address, and blood group.
5. Receptionist links the patient to an existing family or creates a new family.
6. Receptionist submits the form.
7. System generates a unique Patient ID and saves the record.
8. System displays a confirmation with the new Patient ID.

Alternate / Exception Flows

- If the NIC already exists, the system blocks submission and displays the existing patient's summary for the receptionist to verify.
- If the patient is a minor without their own NIC, the receptionist uses the guardian NIC + date-of-birth alternate key.
- If mandatory fields are incomplete, the system highlights the missing fields and prevents submission.

Postconditions	A new patient record exists with a unique Patient ID, linked to a family, and searchable system-wide.
Related Requirements	FR-006 to FR-014

UC-02: Book an Appointment and Manage the Queue

Actors	Receptionist, Doctor
Preconditions	Patient is registered in the system.
Trigger	Patient requests an appointment or arrives as a walk-in.

Main Success Scenario

1. Receptionist searches for the patient by NIC, name, or Patient ID.
2. Receptionist selects an available doctor and time slot (or 'Walk-in — add to today's queue').
3. System validates there is no conflicting booking for that doctor/slot.

4. System creates the appointment with status Waiting and adds it to the live queue.
5. Doctor, from their console, selects 'Call Next Patient'.
6. System updates the appointment status to Called, then to Consulting once the doctor opens the consultation screen.
7. On completion of the consultation, status updates to Completed and the receptionist is notified for billing.

Alternate / Exception Flows

- If the selected slot conflicts with an existing booking, the system rejects the booking and suggests the next available slot.
- If a patient does not respond when Called, the doctor may mark them Skipped and call the next patient; the skipped patient re-enters the queue.

Postconditions	The queue accurately reflects real-time patient status for reception, doctor and (optionally) a waiting-room display.
Related Requirements	FR-024 to FR-031

UC-03: Conduct a Consultation and Issue a Prescription

Actors	Doctor
Preconditions	Patient status is Consulting; doctor has opened the patient's consultation screen.
Trigger	Doctor begins examining the patient.

Main Success Scenario

1. System displays the patient's history, allergies, and chronic conditions.
2. Doctor records vitals (BP, temperature, pulse, weight, height); BMI is auto-calculated.
3. Doctor enters diagnosis and clinical notes.
4. Doctor searches the medicine catalog and adds items to the prescription with dosage, frequency and duration.
5. System validates real-time stock availability for each prescribed item.
6. Doctor finalizes and submits the prescription.
7. System generates a printable prescription and routes it to the Pharmacy Queue with status Pending.
8. System marks the consultation and queue status as Completed.

Alternate / Exception Flows

- If a prescribed medicine conflicts with a documented allergy, the system displays a warning requiring doctor acknowledgement before proceeding.
- If a prescribed item has insufficient stock, the system flags the item; the doctor may substitute, reduce quantity, or proceed with the item flagged for pharmacist follow-up.
- If the doctor is interrupted, the consultation may be saved as a draft and resumed later.

Postconditions	A finalized, timestamped consultation record and an associated prescription exist; the prescription is visible in the pharmacy queue.

Related Requirements	FR-032 to FR-046

UC-04: Dispense a Prescription at the Pharmacy

Actors	Pharmacist
Preconditions	A prescription with status Pending exists in the Pharmacy Queue.
Trigger	Pharmacist selects the next prescription in the queue.

Main Success Scenario

1. Pharmacist opens the prescription; status updates to Preparing.
2. For each item, system suggests the batch to allocate using FEFO logic.
3. Pharmacist confirms the batch and quantity to dispense per item.
4. System validates the batch is not expired and stock is sufficient.
5. Pharmacist confirms dispensing; system deducts the quantity from the batch and updates status to Dispensed.
6. System checks whether remaining stock has fallen below the reorder level and raises a low-stock notification if so.
7. Pharmacist hands over the medicine to the patient and marks status Collected.
8. System appends the dispensed items as billing line items on the visit invoice.

Alternate / Exception Flows

- If the suggested batch is expired or insufficient, the system blocks the Dispensed transition and prompts for an alternate batch or doctor consultation.
- If only some items can be dispensed, the pharmacist may partially dispense, leaving remaining items Pending.
- If the pharmacist manually overrides FEFO allocation, the system requires a documented reason.

Postconditions	Prescribed items are dispensed and recorded against a specific batch; inventory and billing are automatically updated.
Related Requirements	FR-047 to FR-063

UC-05: Manage Inventory, Suppliers and Purchase Orders

Actors	Pharmacist, Administrator
Preconditions	Pharmacist/Administrator is logged in; medicine master data exists.
Trigger	Stock for a medicine approaches or falls below the reorder level, or a scheduled stock review occurs.

Main Success Scenario

1. System raises a low-stock notification on the pharmacist dashboard.
2. Pharmacist reviews suggested reorder quantities and selects a supplier.
3. Pharmacist creates a purchase order listing medicines and quantities; status is set to Draft, then Submitted.

4. Supplier delivers goods; pharmacist records a Goods Received Note against the PO.
5. System creates new stock batches from the GRN, capturing batch number and expiry date.
6. System updates the PO status to Received or Partially Received based on quantities recorded.
7. System closes the PO once all line items are fully received.

Alternate / Exception Flows

- If received quantities differ from ordered quantities, the system flags the discrepancy for Administrator review before closing the PO.
- If a delivered batch is damaged or rejected, the pharmacist records a rejection reason and excludes it from usable stock.

Postconditions	Stock levels and batch records are accurately updated; a complete procurement audit trail exists from PO to GRN to batch.
Related Requirements	FR-064 to FR-078

UC-06: Generate Invoice and Record Payment

Actors	Receptionist, Pharmacist
Preconditions	Consultation status is Completed; any dispensed pharmacy items are recorded.
Trigger	Patient proceeds to settle payment after consultation and/or medicine collection.

Main Success Scenario

1. System auto-generates a consolidated invoice combining the consultation fee and dispensed medicine charges.
2. Receptionist applies any applicable discount (e.g., senior citizen) if configured.
3. Receptionist selects payment method(s) and records the amount(s) received.
4. System confirms full payment and marks the invoice as Paid.
5. System generates a printable/PDF receipt for the patient.

Alternate / Exception Flows

- If the patient pays only part of the amount, the system records a partial payment and keeps the invoice status Outstanding.
- If the invoice requires voiding due to an error, an Administrator must approve the void with a documented reason before the invoice is reversed.

Postconditions	A finalized invoice and payment record exist, linked to the patient's history timeline and the daily billing report.
Related Requirements	FR-079 to FR-085

12. User Stories

User stories translate the use cases above into development-ready backlog items in the standard 'As a ... I want ... so that ...' format, each with brief acceptance criteria.

ID	User Story	Acceptance Criteria (summary)
US-01	As a Receptionist, I want to be warned immediately if a NIC already exists, so that I never create duplicate patient records.	Duplicate NIC entry is blocked with an on-screen match to the existing record.
US-02	As a Receptionist, I want to add a walk-in patient directly to today's queue, so that I don't need to create a formal appointment slot.	Walk-in patient appears in the queue with status Waiting within 2 seconds.
US-03	As a Doctor, I want to see a patient's allergies before prescribing, so that I avoid prescribing a conflicting medicine.	Allergy banner is visible on the consultation screen; a conflicting medicine triggers a blocking warning.
US-04	As a Doctor, I want real-time stock validation while prescribing, so that I don't prescribe medicine that isn't available.	Each prescription line shows live stock status before submission.
US-05	As a Pharmacist, I want the system to suggest the earliest-expiring batch, so that I dispense using FEFO by default.	Suggested batch is always the earliest expiry among valid batches for the medicine.
US-06	As a Pharmacist, I want to be notified when stock drops below the reorder level, so that I can raise a purchase order in time.	Low-stock notification appears on the dashboard within 1 minute of the deduction that caused the breach.
US-07	As an Administrator, I want a daily dashboard of revenue and patient volume, so that I can monitor clinic performance.	Dashboard reflects same-day figures and is exportable to PDF/Excel.
US-08	As a Receptionist, I want a single consolidated invoice per visit, so that patients don't receive separate consultation and pharmacy bills.	One invoice number covers all consultation and dispensed-item charges for a visit.
US-09	As an Administrator, I want every record change logged with user and timestamp, so that I can audit any disputed transaction.	Audit log entry exists for every create/update/delete with before/after values where applicable.
US-10	As a Doctor, I want to view a patient's full visit history on one timeline, so that I can make informed clinical decisions quickly.	Timeline loads all appointments, consultations, prescriptions and invoices in chronological order within 2 seconds.
US-11	As a Pharmacist, I want to record a Goods Received Note against a purchase order, so that new stock batches are created automatically.	GRN submission creates one batch record per line item with correct expiry and quantity.
US-12	As an Administrator, I want to configure reorder levels per medicine, so that low-stock alerts match our actual purchasing needs.	Reorder level is editable per medicine and immediately affects alert thresholds.

13. Activity Diagrams

The following activity flows illustrate the control flow of the two most critical end-to-end processes: the complete patient visit, and the pharmacy dispensing/inventory update sub-process. They are presented here as structured step/decision flows equivalent to the original box-and-arrow diagrams.

Activity Flow: End-to-End Patient Visit Workflow

- ▶ Patient arrives at reception
- ◆ *Existing patient (NIC match)?*
 - Yes → Retrieve patient record
 - No → Register new patient (FR2) + assign Family (FR3)
- ▶ Book / update appointment (FR4)
- ▶ Add patient to queue: Waiting
- ▶ Doctor calls patient → Called
- ▶ Consultation begins → Consulting
- ▶ Doctor records vitals, history, allergies & notes (FR5)
- ◆ *Prescription required?*
 - Yes → Issue e-prescription with medicine lookup (FR6)
 - No → Close consultation record
- ◆ *Stock sufficient?*
 - Yes → Consultation status → Completed
 - No → Flag substitute / notify doctor & patient
- ▶ Route prescription to Pharmacy Queue: Pending (FR7)
- ▶ Pharmacist prepares & dispenses (Preparing → Dispensed)
- ▶ Inventory auto-updates (FR8); status → Collected
- ▶ Generate invoice & record payment (FR10)

Figure 1 — End-to-end activity flow from patient arrival through registration, queueing, consultation, prescription, dispensing and billing.

Activity Flow: Pharmacy Dispensing & Inventory Update Workflow

- ▶ Prescription enters Pharmacy Queue: Pending
- ▶ Pharmacist opens prescription, status → Preparing
- ◆ *Batch available & not expired (FEFO)?*
 - Yes → Select earliest-expiry batch (FEFO allocation)
 - No → Block dispensing; notify pharmacist/doctor
- ▶ Deduct dispensed quantity from batch stock
- ◆ *Stock below reorder level?*
 - Yes → Trigger low-stock notification (FR8)
 - No → Mark status: Dispensed
- ▶ Mark status: Dispensed
- ▶ Patient collects medicine; status → Collected
- ▶ Update inventory ledger & audit log
- ▶ Generate/append billing line item (FR10)

Figure 2 — Pharmacy dispensing workflow showing FEFO batch allocation, stock deduction, low-stock alerting, and billing hand-off.

14. Sequence Diagrams

The sequence flows below show the message exchange between actors and system components for the two core transactions: consultation with prescription issuance, and pharmacy dispensing.

Sequence: Doctor Consultation & E-Prescription Issuance

- ▶ 1. Receptionist → Doctor UI: Call next patient (Queue: Called)
- ▶ 2. Doctor UI → App Server: GET /consultations/{patientId}/history
- ▶ 3. App Server → Patient DB: Fetch history, allergies, vitals
- ▶ 4. Patient DB → App Server: Return patient record
- ▶ 5. App Server → Doctor UI: Render consultation form
- ▶ 6. Doctor UI → App Server: POST /consultations (notes, vitals)
- ▶ 7. App Server → Doctor UI: Show medicine lookup
- ▶ 8. Doctor UI → App Server: POST /prescriptions (items)
- ▶ 9. App Server → Patient DB: Validate stock per item
- ▶ 10. Patient DB → App Server: Stock status
- ▶ 11. App Server → Pharmacy Queue: Push prescription (status: Pending)
- ▶ 12. App Server → Doctor UI: Confirm prescription issued
- ▶ 13. Doctor UI → Receptionist: Update queue status → Completed

Figure 3 — Interaction sequence between Receptionist, Doctor UI, Application Server, Patient Database and Pharmacy Queue during consultation and prescription issuance.

Sequence: Pharmacy Dispensing Process

- ▶ 1. Pharmacist UI → App Server: Open next Pending prescription
- ▶ 2. App Server → Inventory Service: GET available batches (FEFO order)
- ▶ 3. Inventory Service → App Server: Return batch list
- ▶ 4. App Server → Pharmacist UI: Display batch & expiry to pharmacist
- ▶ 5. Pharmacist UI → App Server: Confirm dispense quantity
- ▶ 6. App Server → Inventory Service: POST /inventory/deduct
- ▶ 7. Inventory Service → Inventory Service: Check reorder threshold
- ▶ 8. Inventory Service → App Server: Deduction confirmed
- ▶ 9. App Server → Billing Service: POST /billing/line-item
- ▶ 10. Billing Service → App Server: Invoice line created
- ▶ 11. App Server → Pharmacist UI: Update status → Dispensed / Collected

Figure 4 — Interaction sequence between Pharmacist UI, Application Server, Inventory Service and Billing Service during dispensing.

15. Entity-Relationship (ER) Diagram

The relationship summary below models the core persistent entities of the system and their relationships, forming the basis for the database dictionary in Section 16. Full field-level detail for every entity is given there.

Entity	Related To (Cardinality)
Family	has many → Patient (1:N); one Patient designated head_patient_id
Patient	belongs to → Family (N:1); books → Appointment (1:N)
Appointment	generates → Consultation (1:1)

Entity	Related To (Cardinality)
Consultation	produces → Prescription (1:N)
Prescription	contains → PrescriptionItem (1:N); billed on → Invoice (1:1 optional)
PrescriptionItem	references → Medicine (N:1)
Medicine	stocked as → Batch (1:N)
Batch	supplied via → Supplier (N:1)
Supplier	raised to by → PurchaseOrder (1:N)
PurchaseOrder	created by → User/Staff (N:1)
Invoice	billed for → Patient (N:1)
User (Staff)	logged in → AuditLog (1:N)

Figure 5 — Entity-Relationship model covering patient/family, clinical, pharmacy, procurement, billing and administrative domains.

16. Database Dictionary

The data dictionary below defines each core table, its fields, data types, and constraints, corresponding to the ER relationships in Section 15. This schema is implemented once in Prisma and shared by both the SQLite (local) and PostgreSQL (cloud) targets described in Section 10.3.

Family

Field	Type	Constraints	Description
family_id	BIGINT	PK, AUTO_INCREMENT	Unique family identifier
family_name	VARCHAR(120)	NOT NULL	Household / family display name
head_patient_id	BIGINT	FK → Patient.patient_id, NULL	Designated head of family
address	VARCHAR(255)	NULL	Family home address
contact_no	VARCHAR(20)	NULL	Primary contact phone number
created_at	TIMESTAMP	NOT NULL, DEFAULT now()	Record creation timestamp

Patient

Field	Type	Constraints	Description
patient_id	VARCHAR(12)	PK	System-generated ID, e.g., PT-000123
family_id	BIGINT	FK → Family.family_id, NOT NULL	Family the patient belongs to
nic	VARCHAR(20)	UNIQUE, NULL for minors	National Identity Card number
guardian_nic	VARCHAR(20)	NULL	Used as alternate key for minors
full_name	VARCHAR(150)	NOT NULL	Patient full name

Field	Type	Constraints	Description
dob	DATE	NOT NULL	Date of birth
gender	ENUM	NOT NULL	Male / Female / Other
phone	VARCHAR(20)	NULL	Contact phone number
blood_group	VARCHAR(5)	NULL	e.g., O+, AB-
allergies	TEXT	NULL	Documented known allergies
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Soft-delete / archival flag

Appointment

Field	Type	Constraints	Description
appointment_id	BIGINT	PK, AUTO_INCREMENT	Unique appointment identifier
patient_id	VARCHAR(12)	FK → Patient.patient_id, NOT NULL	Patient booked
doctor_id	BIGINT	FK → User.user_id, NOT NULL	Doctor assigned
scheduled_at	DATETIME	NOT NULL	Slot date/time (null for pure walk-ins)
status	ENUM	NOT NULL, DEFAULT 'Waiting'	Waiting / Called / Consulting / Completed / Skipped
created_by	BIGINT	FK → User.user_id	Receptionist who created the record

Consultation

Field	Type	Constraints	Description
consultation_id	BIGINT	PK, AUTO_INCREMENT	Unique consultation identifier
appointment_id	BIGINT	FK → Appointment.appointment_id, NOT NULL	Related appointment
vitals_json	JSON	NULL	BP, temperature, pulse, weight, height, BMI
allergies_ack	BOOLEAN	DEFAULT FALSE	Doctor acknowledged allergy warning
diagnosis	TEXT	NULL	Diagnosis / assessment
notes	TEXT	NULL	Free-text clinical notes
status	ENUM	NOT NULL, DEFAULT 'Draft'	Draft / Finalized
created_at	TIMESTAMP	NOT NULL, DEFAULT now()	Record creation timestamp

Prescription

Field	Type	Constraints	Description
prescription_id	BIGINT	PK, AUTO_INCREMENT	Unique prescription identifier
consultation_id	BIGINT	FK → Consultation.consultation_id, NOT NULL	Source consultation
status	ENUM	NOT NULL, DEFAULT 'Pending'	Pending / Preparing / Dispensed / Collected
issued_at	TIMESTAMP	NOT NULL, DEFAULT now()	Issuance timestamp

PrescriptionItem

Field	Type	Constraints	Description
rx_item_id	BIGINT	PK, AUTO_INCREMENT	Unique line item identifier
prescription_id	BIGINT	FK → Prescription.prescription_id, NOT NULL	Parent prescription
medicine_id	BIGINT	FK → Medicine.medicine_id, NOT NULL	Prescribed medicine
dosage	VARCHAR(80)	NOT NULL	e.g., '500mg twice daily'
qty	INT	NOT NULL, CHECK > 0	Quantity prescribed
batch_id	BIGINT	FK → Batch.batch_id, NULL until dispensed	Allocated batch

Medicine

Field	Type	Constraints	Description
medicine_id	BIGINT	PK, AUTO_INCREMENT	Unique medicine identifier
name	VARCHAR(150)	NOT NULL	Brand / display name
generic_name	VARCHAR(150)	NULL	Generic/INN name
form	VARCHAR(40)	NULL	Tablet / Syrup / Injection, etc.
unit	VARCHAR(20)	NOT NULL	Unit of measure
reorder_level	INT	NOT NULL, DEFAULT 0	Threshold for low-stock alert
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Discontinued flag

Batch

Field	Type	Constraints	Description
batch_id	BIGINT	PK, AUTO_INCREMENT	Unique batch identifier
medicine_id	BIGINT	FK →	Related medicine

Field	Type	Constraints	Description
		Medicine.medicine_id, NOT NULL	
batch_no	VARCHAR(60)	NOT NULL	Manufacturer batch/lot number
expiry_date	DATE	NOT NULL	Expiry date, used for FEFO ordering
qty_on_hand	INT	NOT NULL, CHECK >= 0	Current available quantity
supplier_id	BIGINT	FK → Supplier.supplier_id, NULL	Source supplier

Supplier

Field	Type	Constraints	Description
supplier_id	BIGINT	PK, AUTO_INCREMENT	Unique supplier identifier
name	VARCHAR(150)	NOT NULL	Supplier company name
contact	VARCHAR(100)	NULL	Primary contact person/number
address	VARCHAR(255)	NULL	Supplier address

PurchaseOrder

Field	Type	Constraints	Description
po_id	BIGINT	PK, AUTO_INCREMENT	Unique purchase order identifier
supplier_id	BIGINT	FK → Supplier.supplier_id, NOT NULL	Ordering supplier
order_date	DATE	NOT NULL	Date order was raised
status	ENUM	NOT NULL, DEFAULT 'Draft'	Draft / Submitted / Partially Received / Received / Closed
created_by	BIGINT	FK → User.user_id	Staff member who created the PO

Invoice

Field	Type	Constraints	Description
invoice_id	BIGINT	PK, AUTO_INCREMENT	Unique invoice identifier
patient_id	VARCHAR(12)	FK → Patient.patient_id, NOT NULL	Billed patient
prescription_id	BIGINT	FK → Prescription.prescription_id, NULL	Related prescription, if any
total_amount	DECIMAL(10,2)	NOT NULL	Total invoice amount
payment_status	ENUM	NOT NULL, DEFAULT	Outstanding / Paid / Voided

Field	Type	Constraints	Description
		'Outstanding'	
created_at	TIMESTAMP	NOT NULL, DEFAULT now()	Invoice creation timestamp

User (Staff)

Field	Type	Constraints	Description
user_id	BIGINT	PK, AUTO_INCREMENT	Unique staff user identifier
role	ENUM	NOT NULL	Admin / Receptionist / Doctor / Pharmacist
username	VARCHAR(60)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	Salted hash of password
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Account enabled flag

AuditLog

Field	Type	Constraints	Description
log_id	BIGINT	PK, AUTO_INCREMENT	Unique audit entry identifier
user_id	BIGINT	FK → User.user_id, NOT NULL	Acting user
action	VARCHAR(40)	NOT NULL	CREATE / UPDATE / DELETE / LOGIN, etc.
entity	VARCHAR(60)	NOT NULL	Affected table/entity name
entity_id	VARCHAR(40)	NOT NULL	Affected record identifier
timestamp	TIMESTAMP	NOT NULL, DEFAULT now()	When the action occurred

17. UI Requirements

The user interface shall be delivered as a responsive React single-page application per Section 10, consuming the Express REST/WebSocket API. The following screens are required as a minimum, organized by module:

Screen	Primary Users	Key Elements
Login / 2FA	All	Username/password fields, 'forgot password' link, optional OTP entry
Dashboard (role-specific)	All	KPI cards, charts, alerts (low-stock, expiry, queue length)
Patient Registration	Receptionist	NIC live-validation field, demographic form, family picker/creator
Family Management	Receptionist	Family list, member roster, head-of-family selector
Appointment & Queue Board	Receptionist, Doctor	Calendar/slot picker, live queue list with status chips, call-next button
Consultation	Doctor	History panel, vitals form, notes editor, allergy

Screen	Primary Users	Key Elements
Workspace		banner, prescription builder
Prescription Builder	Doctor	Medicine search-as-you-type, dosage fields, stock-status indicator per line
Pharmacy Queue	Pharmacist	Prescription list, status Kanban (Pending/Preparing/Dispensed/Collected)
Dispensing Screen	Pharmacist	Batch selector (FEFO-sorted), quantity confirm, barcode scan input
Inventory Management	Pharmacist, Admin	Medicine catalog, batch table, expiry/low-stock filters
Supplier & Purchase Orders	Pharmacist, Admin	Supplier list, PO builder, GRN entry form
Billing / Invoice	Receptionist	Line-item summary, discount entry, payment method selector, print receipt
Patient History Timeline	Doctor, Admin	Chronological feed with filter chips, export-to-PDF button
Reports & Analytics	Admin, Doctor, Pharmacist	Filterable tables/charts, PDF/Excel export buttons
User & Role Management	Admin	User list, role assignment, activate/deactivate toggle
Audit Log Viewer	Admin	Searchable/filterable log table with drill-down detail
System Settings	Admin	Clinic profile, fee schedule, reorder levels, backup status

17.1. General UI Guidelines

- Consistent navigation: a persistent left-hand module menu and top-bar showing logged-in user/role.
- Form validation errors shall be shown inline, adjacent to the offending field, in real time where feasible.
- Destructive actions (delete, void invoice) require a confirmation dialog stating the consequence.
- All list views shall support pagination, sorting, and free-text search.
- Color coding shall follow a consistent convention: green = success/available, amber = warning/low-stock or nearing expiry, red = blocked/expired/error.

18. API Requirements

The system shall expose a versioned, documented RESTful JSON API secured with token-based authentication (JWT via Passport.js, per Section 10.4). All endpoints require an authenticated session and enforce the role permissions defined in Section 6. Representative endpoints per module are listed below.

Authentication

Method	Endpoint	Description
POST	/api/v1/auth/login	Authenticate user and issue access/refresh tokens
POST	/api/v1/auth/logout	Invalidate the current session token
POST	/api/v1/auth/forgot-password	Trigger a password-reset email

Patients & Families

Method	Endpoint	Description
GET	/api/v1/patients?search=	Search patients by NIC, name, phone, or Patient ID
POST	/api/v1/patients	Register a new patient (validates NIC uniqueness)
PUT	/api/v1/patients/{patientId}	Update patient demographic details
GET	/api/v1/families/{familyId}	Retrieve a family and its member roster

Appointments & Queue

Method	Endpoint	Description
POST	/api/v1/appointments	Book a new appointment or walk-in queue entry
PATCH	/api/v1/appointments/{id}/status	Transition queue status (Waiting/Called/Consulting/Completed)
GET	/api/v1/queue/today	Retrieve the live queue for the current day

Consultations & Prescriptions

Method	Endpoint	Description
POST	/api/v1/consultations	Create/finalize a consultation record
GET	/api/v1/patients/{patientId}/history	Retrieve full patient history timeline
POST	/api/v1/prescriptions	Submit a prescription; triggers stock validation
GET	/api/v1/medicines?search=	Search the medicine catalog with live stock summary

Pharmacy & Inventory

Method	Endpoint	Description
GET	/api/v1/pharmacy/queue	Retrieve pending/preparing prescriptions
POST	/api/v1/pharmacy/dispense	Confirm dispensing of a prescription/line item
GET	/api/v1/inventory/batches?medicineId=	Retrieve batches for a medicine, FEFO-sorted
GET	/api/v1/inventory/alerts	Retrieve low-stock and expiry alerts

Suppliers & Purchase Orders

Method	Endpoint	Description
POST	/api/v1/purchase-orders	Create a purchase order
POST	/api/v1/purchase-orders/{id}/grn	Record a Goods Received Note against a PO

Billing

Method	Endpoint	Description
POST	/api/v1/invoices	Generate a consolidated invoice for a visit
POST	/api/v1/invoices/{id}/payments	Record a payment against an invoice
POST	/api/v1/invoices/{id}/void	Void an invoice (requires Admin approval)

Reports & Admin

Method	Endpoint	Description
GET	/api/v1/reports/dashboard?role=	Retrieve role-specific dashboard metrics
GET	/api/v1/reports/export?type=&format=pdf xlsx	Export a report to PDF or Excel
GET	/api/v1/audit-logs?filter=	Retrieve audit log entries

All list endpoints support *page*, *pageSize*, *sort*, and *search query* parameters. All write endpoints return the created/updated resource and a correlation ID for audit-log cross-referencing.

19. Security Requirements

ID	Requirement
SEC-01	All client-server communication shall be encrypted using HTTPS/TLS 1.2 or higher; HTTP requests shall be redirected to HTTPS.
SEC-02	Passwords shall be stored using a modern salted hash algorithm (bcrypt or argon2) with a minimum work factor reviewed annually.

ID	Requirement
SEC-03	Access tokens (JWT or session tokens) shall expire and require refresh; refresh tokens shall be revocable by an Administrator.
SEC-04	Role-based access control shall be enforced server-side on every API endpoint, not solely in the UI.
SEC-05	Sensitive personally identifiable and health information (PII/PHI) shall be encrypted at rest.
SEC-06	All create, update, and delete actions on clinical or financial data shall be recorded in an immutable, timestamped audit log.
SEC-07	The system shall enforce a configurable password policy (minimum length, complexity) and support periodic forced password rotation for Admin accounts.
SEC-08	The system shall lock accounts after repeated failed login attempts and alert Administrators of suspicious login patterns.
SEC-09	Automated daily backups shall be encrypted and stored separately from the primary production environment.
SEC-10	The system shall validate and sanitize all user input server-side to prevent SQL injection, XSS, and CSRF attacks.
SEC-11	Data exports (PDF/Excel) containing patient data shall be watermarked or logged to trace potential data leakage.
SEC-12	The system shall support configurable data-retention and secure-deletion policies aligned to local health-data regulations.
SEC-13	The application shall set secure HTTP response headers (CSP, HSTS, X-Frame-Options, X-Content-Type-Options) on every response.
SEC-14	API endpoints shall be protected by request-rate limiting to mitigate brute-force and denial-of-service attempts.

20. Acceptance Criteria

The following criteria define 'done' for the major feature areas and shall be validated during User Acceptance Testing (UAT) prior to go-live.

Feature Area	Acceptance Criteria
Patient Registration	A duplicate NIC is rejected 100% of the time with a clear message; a unique Patient ID is generated for every valid new registration.
Queue Management	Queue status transitions strictly follow Waiting → Called → Consulting → Completed and are visible in real time to reception and doctor screens.
Prescription & Stock Validation	No prescription can be finalized referencing an out-of-stock or expired medicine without an explicit, logged override.
Dispensing (FEFO)	For any medicine with multiple valid batches, the system-suggested batch is always the one with the earliest expiry date.
Inventory Alerts	A low-stock notification is generated within 1 minute of a deduction that brings stock below the configured reorder level; an expiry alert is generated for any batch within the configured threshold.

Feature Area	Acceptance Criteria
Billing	Every completed visit with a prescription produces exactly one consolidated invoice; invoice totals reconcile to the sum of consultation fee plus dispensed item charges.
Reports	Dashboard and report figures reconcile exactly with underlying transactional data for the selected date range; export files open correctly in standard PDF/Excel viewers.
Security & Audit	Every create/update/delete of a clinical or financial record produces a corresponding audit log entry with correct user, action, and timestamp; unauthorized role access attempts are blocked and logged.
Performance	Patient and medicine searches return within 2 seconds for a database of at least 50,000 patient records under a load of 50 concurrent users.

21. Future Enhancements

The following capabilities are explicitly out of scope for Phase 1 but have been identified as likely candidates for subsequent releases:

Enhancement	Description	Anticipated Phase
Online Booking	Public-facing portal allowing patients to self-book appointments	Phase 2
WhatsApp / SMS Reminders	Automated appointment and follow-up reminders via messaging channels	Phase 2
Mobile App	Native mobile applications for staff and/or patients	Phase 2/3
AI Decision Support	Clinical decision-support suggestions (e.g., drug interaction checking, diagnosis aids)	Phase 3
Laboratory Integration	Full laboratory test ordering, results capture, and reporting for the reserved Laboratory Staff role	Phase 2
Multi-Branch Support	Consolidated operation and reporting across multiple clinic locations	Phase 3

22. Appendices

Appendix A — Glossary

Term	Definition
FEFO	First-Expiry-First-Out — an inventory allocation method that dispenses the batch with the nearest expiry date first.
NIC	National Identity Card — the primary unique identifier used for patient de-duplication.
RBAC	Role-Based Access Control — restricting system access according to a user's assigned role.
GRN	Goods Received Note — a record confirming receipt of goods against a purchase

Term	Definition
	order.
PO	Purchase Order — a formal request to a supplier to deliver specified medicines/quantities.
SRS	Software Requirements Specification — this document.
UAT	User Acceptance Testing — validation performed by end users prior to go-live.
PII / PHI	Personally Identifiable Information / Protected Health Information.
Batch	A specific manufactured lot of a medicine, identified by a batch number and expiry date.
MoSCoW	Prioritization technique: Must-have, Should-have, Could-have, Won't-have (this release).
ORM	Object-Relational Mapper — a library that maps application objects to database rows (Prisma, in this project).
SSR	Server-Side Rendering — generating HTML views on the server rather than in the browser.

Appendix B — Referenced Standards & Conventions

- IEEE 830-1998 — Recommended Practice for Software Requirements Specifications (structural reference).
- ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering.
- WCAG 2.1 — Web Content Accessibility Guidelines, Level AA (referenced in NFR-16).
- MoSCoW Prioritization — Must/Should/Could/Won't, used throughout Section 7.

Appendix C — Original Requirements Brief (Source Document)

This SRS expands the clinic's original one-page requirements brief, summarized below for traceability. All twelve original functional requirement areas (FR1-FR12) and thirteen suggested modules have been retained and elaborated into the detailed requirements, diagrams, and specifications throughout this document. Section 10 (Technology Stack & Architecture), added in Version 3.0 and revised in Version 4.0 to reflect the React/Express stack, records the confirmed implementation stack.

Original Ref	Elaborated In
FR1 — Authentication & Role-Based Access	Section 7.1, Section 19
FR2 — Patient Registration	Section 7.2, UC-01
FR3 — Family Management	Section 7.3
FR4 — Appointment Scheduling & Queue Management	Section 7.4, UC-02
FR5 — Doctor Consultation	Section 7.5, UC-03
FR6 — Electronic Prescription	Section 7.6, UC-03
FR7 — Pharmacy Queue & Dispensing	Section 7.7, UC-04
FR8 — Drug & Inventory Management	Section 7.8, UC-04, UC-05
FR9 — Supplier & Purchase Order Management	Section 7.9, UC-05

Original Ref	Elaborated In
FR10 — Billing, Invoices & Payments	Section 7.10, UC-06
FR11 — Patient History Timeline	Section 7.11
FR12 — Reports & Analytics Dashboards	Section 7.12
New — Technology Stack & Architecture	Section 10